

Solo E-Commerce Platform -- Project Scope, Features & Requirements

Solo E-Commerce Platform

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1. Document Purpose

This document defines the scope of the Solo E-Commerce platform, the functional requirements (FR-xxx) and **non-functional requirements (NFR-xxx)** that the system must satisfy, the user roles, the assumptions and constraints, the in/out-of-scope items, and the future roadmap. It is a contractual reference for engineering, QA, design and operations. Companions: 01 HLD, 02 LLD, 03 Architecture, 05 Technical Features.

2. Project Vision & Goals

Solo is a multi-brand, multi-category, premium-positioned e-commerce storefront serving the UAE market initially, with planned expansion to KSA. The platform combines:

- A modern, content-rich storefront for browsing and purchase.
- A complete admin/operations console for catalog, orders, content, customers, returns, loyalty and audit.
- A clean, headless API ready for a future mobile companion app (React Native / Capacitor).

Strategic goals:

ID	Goal
G1	Deliver a fast, premium storefront experience (Core Web Vitals "Good" on p75)
G2	Enable non-engineering staff to fully manage catalog and content via the admin SPA
G3	Provide regulator-grade order/payment/audit history (UAE VAT, FTA, PDPL)
G4	Support multiple payment rails: Stripe (cards), Tabby & Tamara (BNPL), Cash on Delivery
G5	Build a loyalty programme that drives repeat purchase
G6	Operate on cloud-native, IaC-managed Azure infrastructure with one-command deploy
G7	Maintain a green, secure baseline against OWASP Top 10
G8	Make the platform observable with end-to-end traces, structured logs, and dashboards

3. Stakeholders & Roles

3.1 External Users

Role	Description
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Guest	Anonymous visitor; can browse, search, build cart, start checkout
Customer	Registered shopper; full purchase, address book, orders, returns, favorites, loyalty
B2B Customer	Bulk-order requests with quote workflow

3.2 Internal Users

Role	Description
Admin	Full back-office access: catalog, orders, customers, content, audit, settings
Super Admin	Admin + system-level settings, role management, irreversible actions

(Future: granular sub-roles such as CONTENT_EDITOR, ORDER_MANAGER, SUPPORT_AGENT -- see roadmap §10.)

3.3 External Systems

System	Role
Stripe	Card payments + webhook
Tabby	BNPL provider
Tamara	BNPL provider
SMTP relay	Transactional email (order confirmation, password reset, etc.)
Azure Blob Storage	Product imagery and CMS media
Application Insights	Telemetry, traces, RUM

4. In Scope

- React 19 + Vite SPA storefront (responsive, mobile-first, light/dark)
- Admin SPA (lazy-loaded, role-gated)
- NestJS REST API (single deploy unit, ~50 modules)
- PostgreSQL data store (~50 tables)
- Stripe + Tabby + Tamara + Cash on Delivery payment rails
- UAE VAT (5%), AED currency
- Loyalty wallet (earn + redeem)
- Promo codes (percentage + fixed)
- Returns workflow (request -> approve -> refund)
- CMS for home page, landing pages, banners, navigation, collections, announcements

- Append-only audit trail across all admin writes
- Azure deployment via azd up / azd deploy with Bicep IaC
- Application Insights + Log Analytics observability
- Email notifications via SMTP

5. Out of Scope (current release)

- Native mobile applications (planned in roadmap)
- Multi-currency at runtime (AED only at launch -- KSA roadmap)
- Multi-language UI beyond English (Arabic in roadmap)
- Marketplace / multi-vendor sellers
- Real-time chat support (use third-party widget for launch)
- Subscription / recurring billing
- Direct social-commerce integrations (Instagram Shop, TikTok Shop)
- ERP / accounting integration (CSV export at launch; deeper roadmap)

- In-store POS integration

6. Functional Requirements

6.1 Storefront -- Catalog & Browse

ID	Requirement
FR-001	Storefront SHALL render a configurable home page assembled from CMS sections (hero banner, category tiles, brand strip, collections, editorial, best sellers, new arrivals, promo strip, testimonials)
FR-002	Storefront SHALL provide category and subcategory landing pages
FR-003	Storefront SHALL provide brand and designer landing pages
FR-004	Storefront SHALL provide a faceted product list with filters: category, brand, designer, price range, in-stock, on-sale, new, featured, best seller
FR-005	Storefront SHALL support sort orders: price asc/desc, newest, popular
FR-006	Storefront SHALL paginate product lists with configurable page size
FR-007	Storefront SHALL provide full-text search over product name, SKU, brand, description
FR-008	Product detail page SHALL display gallery, price (incl + excl VAT), stock, variant axes (color/size/material), description, related products
FR-009	Variants SHALL be addressable by slug; switching a variant SHALL update the URL without full reload
FR-010	Storefront SHALL render schedulable banners and announcements

6.2 Account, Auth & Identity

ID	Requirement
FR-020	Users SHALL register with email + password (Argon2id hashed)
FR-021	Users SHALL receive an email-verification link valid for 1 hour
FR-022	Users SHALL log in and receive a 15-minute access JWT plus a 7-day refresh token
FR-023	Refresh tokens SHALL be rotated on use; reuse SHALL revoke the entire chain
FR-024	Users SHALL be able to request password reset via email link
FR-025	Authenticated users SHALL view & update profile (first/last name, phone)
FR-026	Users SHALL maintain an address book (multiple shipping + billing addresses)
FR-027	Users SHALL view order history with status, totals, and downloadable invoice
FR-028	Users SHALL view loyalty balance, lifetime earn, and transaction ledger
FR-029	Users SHALL maintain a favorites list (idempotent add/remove)
FR-030	Users SHALL request returns from delivered orders

6.3 Cart & Checkout

ID	Requirement
FR-040	Guests and authenticated users SHALL build a cart
FR-041	Cart totals SHALL be computed server-side on every read (never trust client)
FR-042	Adding the same SKU to cart twice SHALL merge quantities
FR-043	Setting line quantity to zero SHALL remove the line
FR-044	Promo codes SHALL be validatable from cart with clear error messaging
FR-045	Loyalty redemption SHALL be capped at $\min(\text{walletBalance}, \text{subtotal} \times \text{maxRedeemPct})$
FR-046	Checkout SHALL collect shipping address, shipping method, and payment method
FR-047	Checkout SHALL show a final total breakdown before payment confirmation
FR-048	On Stripe selection the SPA SHALL open Stripe Elements with clientSecret from API
FR-049	On Tabby/Tamara the SPA SHALL redirect to provider-hosted checkout
FR-050	On Cash on Delivery the order SHALL be created in PROCESSING directly

6.4 Orders, Payments & Returns

ID	Requirement
FR-060	Order creation SHALL atomically lock product rows, snapshot totals, decrement stock, post pending loyalty earn, and write audit
FR-061	Stock reservation SHALL move from reservedQty to deducted on PAID
FR-062	Webhook processing SHALL be idempotent via provider event tables (stripe_events, tabby_events, tamara_events)
FR-063	Invoices SHALL be generated as PDF (pdfkit), stored, and downloadable
FR-064	Order status transitions SHALL follow the documented state machine and emit order_status_history rows
FR-065	Refunds via Stripe SHALL set order status to REFUNDED and roll back loyalty earn
FR-066	Returns SHALL move through REQUESTED -> APPROVED -> IN_TRANSIT -> RECEIVED -> COMPLETED (or REJECTED)
FR-067	Approved returns SHALL trigger Stripe refund + loyalty wallet refund

6.5 Admin -- Catalog & Content

ID	Requirement
FR-080	Admin SHALL create / update / soft-delete / restore products with images, pricing, and flags
FR-081	Admin SHALL manage categories, subcategories, brands, designers (CRUD)
FR-082	Admin SHALL manage product groups (variant sets)
FR-083	Admin SHALL upload images via the media module; sharp SHALL generate thumb/medium/large variants
FR-084	Admin SHALL configure the home page (sections, ordering, scheduling)
FR-085	Admin SHALL CRUD landing pages, banners, navigation, collections, announcements
FR-086	Admin SHALL CRUD promo codes (PERCENT/FIXED, min order, max uses, per-user cap, schedule)

6.6 Admin -- Operations

ID	Requirement
FR-090	Admin SHALL view dashboard KPIs (revenue today/7d, orders today, new customers 7d, low-stock count)
FR-091	Admin SHALL view the orders queue filtered by status, date range, customer
FR-092	Admin SHALL transition orders through the documented state machine
FR-093	Admin SHALL view customer profiles with order history and loyalty balance
FR-094	Admin SHALL adjust loyalty balances (ADJUSTED ledger entry) with reason captured in audit
FR-095	Admin SHALL receive low-stock notifications (in-app + email-optional)
FR-096	Admin SHALL generate reports: sales summary, revenue period, top products, customers LTV, orders by status, low stock
FR-097	Admin SHALL view the audit log filtered by user, action, entity, date range
FR-098	Admin SHALL configure site settings (VAT rate, shipping tiers, loyalty earn rate, redemption cap)
FR-099	Admin SHALL handle bulk-order (B2B) requests through approval workflow

7. Non-Functional Requirements

7.1 Performance

ID	Requirement
NFR-001	API p95 read latency SHALL be <= 700 ms region-internal
NFR-002	API p95 write latency SHALL be <= 1.2 s
NFR-003	LCP on storefront pages SHALL be <= 2.5 s on 3G p75
NFR-004	CLS SHALL be <= 0.1 on storefront pages
NFR-005	Container cold-start SHALL be <= 5 s
NFR-006	Admin lazy chunk SHALL not exceed 500 KB gzipped

7.2 Scalability

ID	Requirement
NFR-010	API SHALL autoscale 1-10 replicas based on concurrent HTTP requests
NFR-011	DB SHALL be vertically scalable without code changes
NFR-012	The system SHALL sustain ≥ 50 concurrent checkouts per minute at launch
NFR-013	Catalog read endpoints SHALL be safe to add a read-replica fan-out later

7.3 Availability

ID	Requirement
NFR-020	Monthly availability SHALL be $\geq 99.9\%$ (≤ 43 min downtime / month)
NFR-021	Planned maintenance SHALL be announced 48 h in advance
NFR-022	Application Insights availability test SHALL ping /api/health every 5 minutes

7.4 Security

ID	Requirement
NFR-030	All transport SHALL be HTTPS with HSTS
NFR-031	Passwords SHALL be hashed with Argon2id (memory ≥ 19 MB, time-cost ≥ 2)
NFR-032	Access JWTs SHALL expire ≤ 15 min; refresh tokens ≤ 7 d, rotated on use
NFR-033	All inputs SHALL pass class-validator whitelist + transform
NFR-034	Helmet, CORS allow-list, ThrottlerGuard SHALL be globally enabled
NFR-035	Webhooks SHALL verify HMAC signatures and dedup via provider event tables
NFR-036	Containers SHALL run as non-root with minimal Node 20 alpine image
NFR-037	Secrets SHALL be sourced from Container App secrets (Key Vault recommended)
NFR-038	The system SHALL maintain an OWASP Top 10 (2021) green checklist

7.5 Compliance

ID	Requirement
NFR-040	Invoices SHALL include UAE TRN, VAT split, sequential invoice number
NFR-041	VAT rate SHALL be snapshotted per order (vatRateSnapshot) for retro correctness
NFR-042	Audit log SHALL be append-only with ≥ 90 -day hot retention and cold export
NFR-043	Personal data deletion requests SHALL be supported (PDPL/GDPR alignment)

7.6 Accessibility

ID	Requirement
NFR-050	Storefront SHALL meet WCAG 2.1 AA on key flows (browse, PDP, cart, checkout, account)
NFR-051	All interactive elements SHALL be keyboard-navigable with visible focus
NFR-052	Forms SHALL have proper label/aria associations

7.7 Maintainability & DevOps

ID	Requirement
NFR-060	Codebase SHALL be 100% TypeScript across both tiers
NFR-061	All endpoints SHALL be documented in Swagger UI at /api/docs
NFR-062	Backend test coverage on services SHALL be $\geq 70\%$
NFR-063	Frontend test coverage on stores + hooks SHALL be $\geq 70\%$
NFR-064	CI SHALL run lint + test + build on every PR
NFR-065	One-command production deploy SHALL be supported via azd deploy

7.8 Observability

ID	Requirement
NFR-070	Backend SHALL emit Pino structured logs with correlation IDs
NFR-071	Backend SHALL emit App Insights traces for every HTTP request
NFR-072	Custom events SHALL be emitted for ORDER_PAID, ORDER_REFUNDED, RETURN_COMPLETED
NFR-073	Alerts SHALL fire on error-rate > 1% / 5 min, container-restart > 3 / 10 min, PG CPU > 80% / 10 min

8. Assumptions & Constraints

#	Item
A1	Single primary region: Azure East US 2 (low-cost; UAE Central optional)
A2	Single currency at launch: AED
A3	Single language at launch: English (LTR)
A4	Stripe, Tabby, Tamara provider accounts available
A5	SMTP relay credentials available
A6	UAE TRN issued for invoice generation
C1	Budget excludes premium WAF in launch (recommended in roadmap)
C2	Mobile native app deferred to roadmap
C3	Marketplace / multi-vendor model not in scope
C4	No integration with external ERP at launch (CSV export only)

9. Acceptance Criteria

The release SHALL be considered ready when:

- All FR-0xx requirements are implemented and covered by automated tests.
- All NFR-0xx requirements are measured and met in pre-prod load tests.
- Penetration test (OWASP scope) yields no Critical or High findings.
- Disaster recovery drill (DB restore + frontend redeploy) completes

inside the 4-hour RTO target.

- Admin user-acceptance test sign-off on all admin workflows.
- Customer user-acceptance test sign-off on browse/PDP/cart/checkout/account/returns flows.

10. Future Roadmap

Phase	Feature
Phase 2 (Q3 2026)	Arabic locale (RTL), KSA market launch (15% VAT), multi-currency
Phase 2	Granular admin sub-roles (CONTENT_EDITOR, ORDER_MANAGER, SUPPORT_AGENT)
Phase 3 (Q4 2026)	Mobile companion app -- React Native (Expo) or Capacitor wrapper sharing the React component library
Phase 3	Real-time inventory sync from supplier feeds
Phase 4 (Q1 2027)	Multi-region active-passive (Azure UAE Central + East US 2)
Phase 4	Azure Front Door + WAF in front of SWA + ACA
Phase 4	Dedicated Redis (cart sessions, rate-limit counters, hot caches)
Phase 5	Personalised recommendations (Azure ML)
Phase 5	Marketplace / multi-vendor support
Phase 5	ERP & accounting integration (Microsoft Dynamics, SAP B1)

- > Note: The mobile app strategy is React Native or Capacitor
- > -- not Flutter -- so we can share components, types, and the existing
- > API client with the web codebase.

End of Project Scope, Features & Requirements.